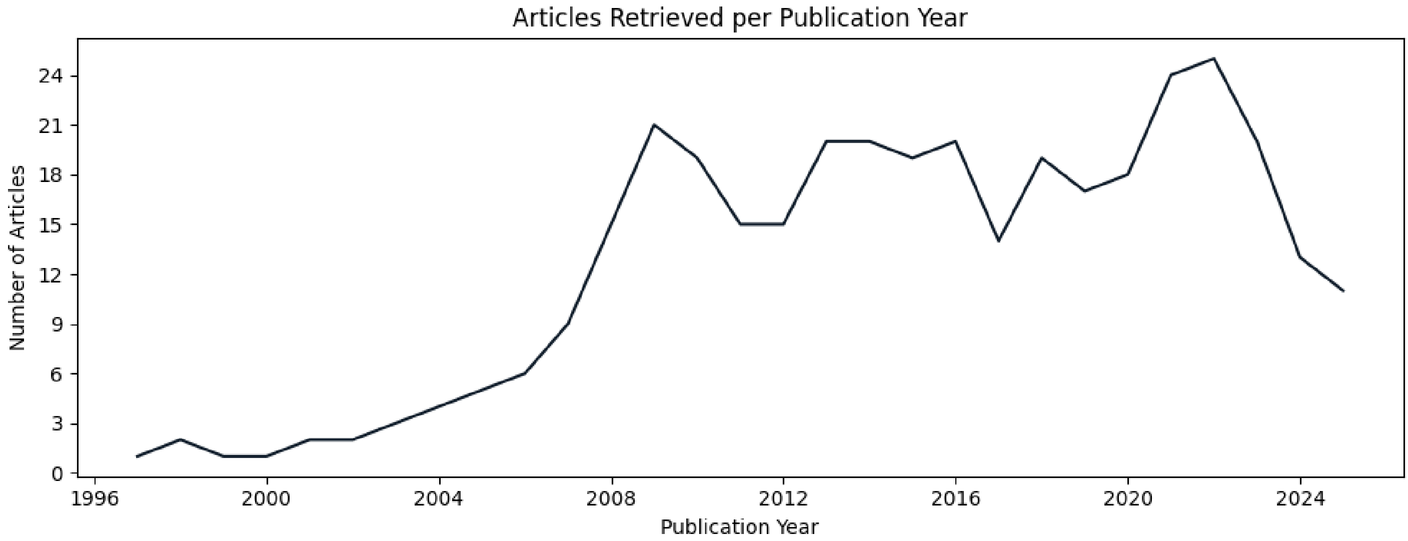
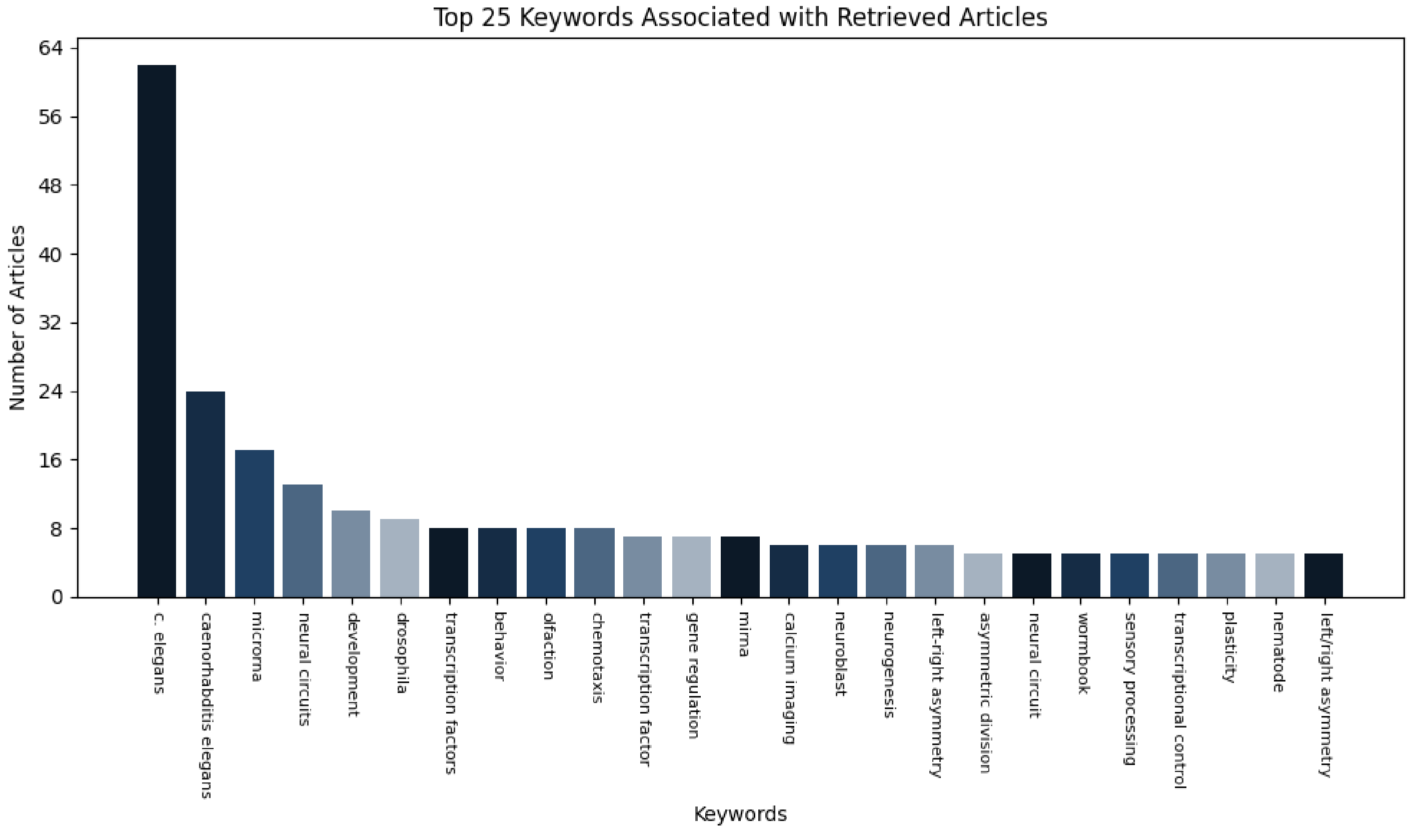
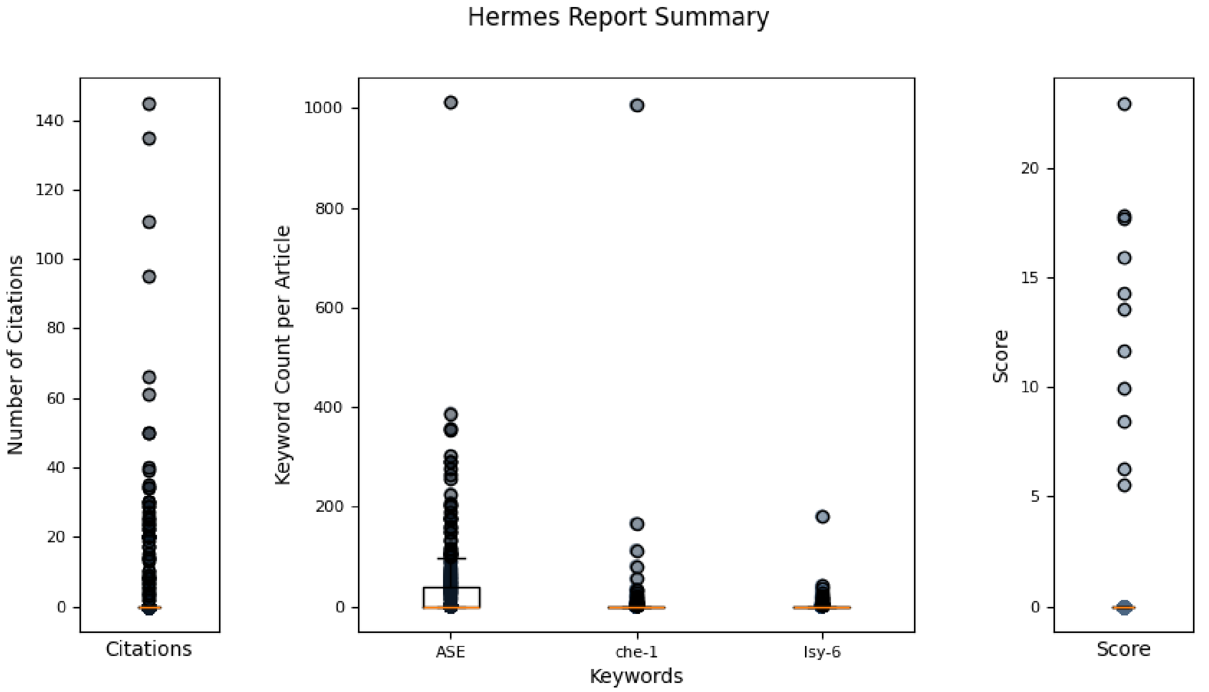
Literature Mining Report

Query Statistics

Query date: 2025-08-25
Query terms: "ASE asymmetry c elegans"
Articles parsed/hits: 364/364 (100.00%)
Scoring keywords: ASE, che-1, lsy-6 (Strict Mode)







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
33002421	2020	Developmental C	Charest, Julien et al.	combinatorial action of temporally segregated...	27	22.96
39167071	2024	Genetics	Poole, Richard et al.	Neurogenesis in Caenorhabditis elegans	8	17.81
40586702	2025	eLife	NA	evolution of lateralized gustation in nematod...	3	17.65
31526477	2019	eLife	Hong, Ray L et al.	evolution of neuronal anatomy and circuitry i...	34	15.91
39282255	2025	bioRxiv	Mackie, Marisa et al.	evolution of lateralized gustation in nematod...	0	14.27
21698137	2011	PLoS Genetics	Poole, Richard et al.	a genome-wide rnai screen for factors involve...	0	13.58
19875417	2010	Nucleic Acids R	Takayama, Jun et al.	single-cell transcriptional analysis of taste...	0	11.62
34504291	2021	Communications	Dekkers, Martij et al.	plasticity in gustatory and nociceptive neuro...	0	9.92
24065887	2013	Frontiers in Ce	Alqadah, Amel et al.	microrna function in left-right neuronal asym...	0	8.43
29481550	2018	Nature methods	Alberti, Chiara et al.	cell-type specific sequencing of micrornas fr...	0	6.28
26635721	2015	Frontiers in Ne	Davis, Gregory et al.	micrornas: not fine-tuners but key regulato...	0	5.57

Query Result 1/11

Combinatorial Action of Temporally Segregated Transcription Factors

Charest, Julien; Daniele, Thomas; Wang, Jingkui; Bykov, Aleksandr; Mandlbauer, Ariane; Asparuhova, Mila; Rhsner, Josef; Gutirrez-Prez, Paula; Cochella, Luisa

Developmental Cell

Year: 2020 PMID: 33002421 doi: 10.1016/j.devcel.2020.09.002 Citations: 27 Score: 22.96

Keywords hits: ASE: 179, che-1: 55, lsy-6: 180

Abstract:

Combinatorial action of transcription factors (TFs) with partially overlapping expression is a widespread strategy to generate novel gene-expression patterns and, thus, cellular diversity. Known mechanisms underlying combinatorial activity require co-expression of TFs within the same cell. Here, we describe the mechanism by which two TFs that are never co-expressed generate a new, intersectional expression pattern in C.elegans embryos: lineage-specific priming of a gene by a transiently expressed TF generates a unique intersection with a second TF acting on the same gene four cell divisions later; the second TF is expressed in multiple cells but only activates transcription in those where priming occurred. Early induction of active transcription is necessary and sufficient to establish a competent state, maintained by broadly expressed regulators in the absence of the initial trigger. We uncover additional cells diversified through this mechanism. Our findings define a mechanism for combinatorial TF activity with important implications for generation of cell-type diversity.

Summary:

Early expression of TBX-37/38 is necessary for lsy-6 transcription in ASEL four cell divisions later. These data indicate that the requirement for TBX-37/38 is mediated through a binding site downstream of the lsy-6 miRNA sequence. It was previously shown that early expression of TBX-37/38 could prime a lsy-6 reporter transgene for subsequent activation by CHE-1, but expression of TBX-37/38 after the birth of the ASE neurons was unable to do so. We generated a transgenic strain carrying lineage-specific reporters, synchronized and dissociated embryos, and sorted viable cells from ABa or ABp based on fluorescence, at different time points. The earliest time point at which we could separate cells from the different lineages was at the onset of TBX-37/38 expression in the ABa lineage. We conclude that early adoption of a transcriptionally active state is necessary and sufficient to prime lsy-6 for later robust activation and explains the requirement for TBX-37/38. Given that TBX-37/38 bind the lsy-6 locus early and are not continuously required, we set out to investigate how this competent state of lsy-6 is maintained. Broadly acting TFs could promote low levels of transcription of the lsy-6 locus that, together with the active chromatin marks, may propagate the competent state until the onset of CHE-1. Priming of enhancers prior to the time point when robust transcription is required is a broadly observed phenomenon that can be caused by different mechanisms, e. g., pre-establishment of contact between promoter and distal enhancers, action of pioneer TFs, acquisition of chromatin modifications, or as shown here by promoting early antisense transcription.

Figures:

Figure 1: Integration of Transcriptional Inputs over Time as a Mechanism for Cell Diversification

Figure 2: Early, Direct Binding of TBX-37/38 tolsy-6Is Necessary and Sufficient to Establish Competence for Transcription in the ASE Neurons

Figure 3: TBX-37/38 Are Not Continuously Required forlsy-6Expression

Figure 4: TBX-37/38 Establish a Differentially Accessible State of thelsy-6Locus

Figure 5: Transcription over thelsy-6Locus Occurs Bidirectionally and Is Required for Priming

Figure 6: Asymmetriclsy-6Competence Is Maintained in a SymmetricTrans-acting Factor Environment

Figure 7: TBX-37/38 Determine Left-Right Molecular Asymmetry across Multiple Bilateral Neuron Pairs with ABa/ABp Lineage Origin

Mentioned biomedical entities:

Genes: ASER, B, CHE-1, Cas9, FoxA, L1, PHA-4, TBX-37, TBX-38, TEL, ZIF-1, Zdbf2, galK, gcy-5, histone, lsy-6, unc-119

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cytosol, neuron, zygote

Organisms: mouse

Tissues: egg, embryo, neural crest

Pathways: N/A

Query Result 2/11

Neurogenesis in Caenorhabditis elegans

Poole, Richard J; Flames, Nuria; Cochella, Luisa; Sundaram, M

Genetics

Year: 2024 PMID: 39167071 doi: 10.1093/genetics/iyae116 Citations: 8 Score: 17.81

Keywords hits: ASE: 149, che-1: 34, lsy-6: 17

Abstract:

Animals rely on their nervous systems to process sensory inputs, integrate these with internal signals, and produce behavioral outputs. This is enabled by the highly specialized morphologies and functions of neurons. Neuronal cells share multiple structural and physiological features, but they also come in a large diversity of types or classes that give the nervous system its broad range of functions and plasticity. This diversity, first recognized over a century ago, spurred classification efforts based on morphology, function, and molecular criteria. Caenorhabditis elegans, with its precisely mapped nervous system at the anatomical level, an extensive molecular description of most of its neurons, and its genetic amenability, has been a prime model for understanding how neurons develop and diversify at a mechanistic level. Here, we review the gene regulatory mechanisms driving neurogenesis and the diversification of neuron classes and subclasses in C. elegans. We discuss our current understanding of the specification of neuronal progenitors and their differentiation in terms of the transcription factors involved and ensuing changes in gene expression and chromatin landscape. The central theme that has emerged is that the identity of a neuron is defined by modules of gene batteries that are under control of parallel yet interconnected regulatory mechanisms. We focus on how, to achieve these terminal identities, cells integrate information along their developmental lineages. Moreover, we discuss how neurons are diversified postembryonically in a time-, genetic sex-, and activity-dependent manner. Finally, we discuss how the understanding of neuronal development can provide insights into the evolution of neuronal diversity.

Summary:

Together, these approaches revealed that combinations of key TFs are necessary for defining neuron class identity and act, for the most part, by directly activating the expression of broad batteries of terminal effector genes. In the section on neuronal diversification, we go deeper into how combinations of terminal selectors achieve neuron class-specific gene expression and how these combinations are generated during development. Mutational analysis of terminal selectors in multiple different contexts has revealed that while most neuron-specific genes fail to be normally expressed, there are at least 2 gene batteries that are typically unaffected: panneuronal genes, which as explained above are primarily regulated by the CUT TFs, and genes coding for structural cilia components, which are shared by all sensory ciliated neurons. Other key regulators of neural precursor specification in both Drosophila and vertebrates include the HOX cluster HD TFs. As discussed below, many but not all these genes have been shown to regulate neural vs nonneural decisions and the specification of neural progenitors. The discovery in Drosophila that achaetescute controls the development of the external sensory organ precursors and that atonal controls the development of the internal chordotonal organs led to the realization that proneural genes not only regulate the specification of neuronal precursors but may also play important roles in the differentiation of different classes of neurons. This classic proneural function is not restricted to postembryonic lineages as lin-32 is also expressed and required in the ABplaaaaa/ABarpapaa neuroblasts that give rise respectively to the left and right URAD, CEPD, and URX neurons. Several other C. elegans proneural genes have also been shown to regulate the specification of neuroblasts including the AS-C homolog hlh-14, which is required for the embryonic specification of the PVQ/HSN/PHB neuroblasts, the ABalpppp/ABpraaap neuroblasts, the DVC neuroblast, and the PVR neuron the NeuroD homolog cnd-1, which is required for the specification of the neuroblasts that give rise to the embryonic DA, DB, and DD ventral nerve cord motor neurons and the Neurogenin homolog ngn-1, which is required for the LR asymmetric specification of the MI neuron.

Figures:

Figure 1: Regulatory framework for C. elegans neurogenesis.

Figure 2: Developmental origin of neurons and neuronal diversity.

Figure 3: Terminal selector functions in neuronal diversification.

Figure 4: Generation of terminal selector combinations in space and time.

Figure 5: Repressor-based mechanisms for neuronal diversification.

Figure 6: Environmental effects on neuron diversity.

Figure 7: Progressive determination model of neuronal development.

Figure 8: Expression of bHLH TFs in neuronal lineages.

Mentioned biomedical entities:

Genes: ASE, AST-1, B, CEH-10, CHE-1, CUT, DAF-19, DDs, HD, Hh, IL2, MEC-3, MyoD, NGF, Neurogenin, Notch, ONECUT, PHB, RIP, Sox, Sox-3, Sox2, SoxB, SoxC, UNC-18, UNC-3, UNC-86, ceh-13, ces-1, ces-2, egl-5, hlh-3, hlh-6, lin-32, lin-39, mab-5, ngn-1, nob-1, php-3, sox-2, unc-3, unc-30, unc-4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cilium, ectoderm, glial cell, mesoderm, muscle cell, neural progenitor cell, neural stem cell, neuron

Organisms: C. elegans, mice, zebrafish

Tissues: hypodermis, nervous system, neural tissue, tissue

Pathways: N/A

Query Result 3/11

Evolution of lateralized gustation in nematodes

NA

eLife

Year: 2025 PMID: 40586702 doi: 10.7554/eLife.103796.3.sa0 Citations: 3 Score: 17.65

Keywords hits: ASE: 177, che-1: 112, lsy-6: 7

Abstract:
Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in Pristionchus pacificus respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode Caenorhabditis elegans. Previously we showed that P. pacificus likely lacked bilateral asymmetry (Hong et al., 2019). Here, we show that by visualizing neuronal activity patterns, contrary to previous expectations based on its genome sequence, the salt responses of P. pacificus are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:
Since NH4I sensation is affected by silencing of che-1 neurons but is unaffected in che-1 mutants, ASE function may be more greatly impacted by the silencing of ASE than by the loss of che-1. To assess whether Ppa ASE neurons show the same lateralized response to salt ions as Cel ASE neurons, we generated transgenic P. pacificus lines that express RCaMP in ASE neurons and assessed calcium responses to attractive salts. Specifically, although we observed weaker or comparable responses in AFD neurons when compared to either ASE neurons response toward 250 mM NH4 Cl, Na Cl, and NH4I, the AFD responses were more robust than ASER toward 25 mM NH4 Cl, Na Cl, and NH4I. However, the responses to 250 mM Na Cl were not significantly reduced in either the ASEL or ASER neuron in the gcy-22.3 mutant. We show that three neuron types each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. RLH, OH, and SHC contributed to the analysis and writing of the work.

Figures:

Figure 1: A comparison of chemotaxis responses to water-soluble ions betweenP. pacificusandC. elegans.

Figure 2: P. pacificus che-1expression in ASE and AFD amphid neurons.

Figure 3: Co-expression ofgcy-22.3p::GFPandttx-1p::RFPin ASER.

Figure 4: che-1expressing amphid neurons are required for sensing water-soluble ions inP. pacificus.

Figure 5: Theche-1p::HisCl1transgene is necessary for histamine-dependent knockdown of salt attraction.

Figure 6: Molecular lesions forP. pacificus che-1andgcy 22.3.

Figure 7: ASEL and ASER responses to different concentrations of NH4Cl, NaCl, and NH4I.

Figure 8: Negative controls of ASE and AFD neuron responses to green light.

Figure 9: Individual traces and heatmaps of ASE responses to various salts.

Figure 10: Microfluidic apparatus used for calcium imaging.

Figure 11: Combined AFD responses in comparison to ASE left or right neuron responses to NH4Cl, NaCl, and NH4I.

Figure 12: Individual traces and heatmaps of AFD responses to various salts.

Figure 13: AFDL/R responses to different concentrations of NH4Cl, NaCl, and NH4I.

Figure 14: The laterally asymmetric expression ofgcy-22.3is dependent on the zinc finger transcription factor CHE-1.

Figure 15: ASE and AFD responses in wildtype compared togcy-22.3mutants.

Figure 16: AFDL and AFDR responses to NH4Cl and NaCl in wildtype andgcy-22.3mutant.

Figure 17: Chemotaxis responses to individual ions ingcy-22.3mutants.

Mentioned biomedical entities:

Genes: ADF, ASER, CHE-1, Cas9, Cel, HisCl1, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, fozi-1, gcy-5, lim-6, lin-4, lsy-6, rGC, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: neuron

Organisms: C. elegans, Pristionchus pacificus, human

Tissues: N/A

Pathways: N/A

Query Result 4/11

Evolution of neuronal anatomy and circuitry in two highly divergent nematode species

Hong, Ray L; Riebesell, Metta; Bumbarger, Daniel J; Cook, Steven J; Carstensen, Heather R; Sarpolaki, Tahmineh; Cochella, Luisa; Castrejon, Jessica; Moreno, Eduardo; Sieriebriennikov, Bogdan; Hobert, Oliver; Sommer, Ralf J; Sengupta, Piali; Marder, Eve; Sengupta, Piali

eLife

Year: 2019 PMID: 31526477 doi: 10.7554/eLife.47155.053 Citations: 34 Score: 15.91

Keywords hits: ASE: 90, che-1: 33, lsy-6: 4

Abstract:
The nematodes C. elegans and P. pacificus populate diverse habitats and display distinct patterns of behavior. To understand how their nervous systems have diverged, we undertook a detailed examination of the neuroanatomy of the chemosensory system of P. pacificus. Using independent features such as cell body position, axon projections and lipophilic dye uptake, we have assigned homologies between the amphid neurons, their first-layer interneurons, and several internal receptor neurons of P. pacificus and C. elegans. We found that neuronal number and soma position are highly conserved. However, the morphological elaborations of several amphid cilia are different between them, most notably in the absence of winged cilia morphology in P. pacificus. We established a synaptic wiring diagram of amphid sensory neurons and amphid interneurons in P. pacificus and found striking patterns of conservation and divergence in connectivity relative to C. elegans, but very little changes in relative neighborhood of neuronal processes. These findings demonstrate the existence of several constraints in patterning the nervous system and suggest that major substrates for evolutionary novelty lie in the alterations of dendritic structures and synaptic connectivity.

Summary:
All amphid neurons and the amphid sheath cell in P. pacificus have their cell bodies in the lateral ganglion posterior to the nerve ring, which resembles the condition found in C. elegans and other nematodes. Similar to AFD endings in other species, the AM12 cilium is fully embedded in the amphid sheath cell process, does not enter the lumen of the sheath cell or the amphid channel and thus has no contact to the outside environment. As the AM7 and AWC axons both cross the dorsal midline and terminate above the lateral midline, we regard them as homologs based on this singular feature, although the AM7 cell body appears to have shifted from a position ventral of ASH in C. elegans to a position between AM1 and AM8 in P. pacificus, such that the three cell types are just ventral and parallel to the lateral midline on each side. Two other amphid neurons also show strong resemblance to their C. elegans counterparts according to their unique axon projections. Four of the strong gap junction connections were present in both species while 6 and 2 were specific to P. pacificus and C. elegans, respectively. Hence, like odr-7 expression, the conservation of cell-type specific expression is limited to the amphid neurons but not to the proposed cellular homologs when soma position and axon projection patterns are also considered. In the absence of morphological characteristics such as the winged endings of AWx neurons in C. elegans, we proposed neuronal homologies between P. pacificus and C. elegans amphid neurons based on their relative cell body positions, axon projections, the manner of dendrite entry into the amphid sheath cell, the number of cilia in channel, dye filling properties, and the connections to first layer interneurons.

Figures:

Figure 1: EM of the amphid sensillum and two other sensory neurons ofP. pacificushermaphrodite adults.

Figure 2: URX is ciliated.

Figure 3: URY, URX, URA and BAG endings.

Figure 4: A comparison of select non-amphid ciliated neurons in three nematode species.

Figure 5: Video of URX 3D reconstruction.

Figure 6: Video of URY 3D reconstruction.

Figure 7: Video of URA 3D reconstruction.

Figure 8: Position of amphid cilia in AMso and AMsh.

Figure 9: Posterior-anterior path of amphid dendrite entry into the left sheath glia of specimen 107.

Figure 10: Posterior-anterior path of amphid dendrite entry into the right sheath glia of specimen 107.

Figure 11: Overview of the amphid sensilla.

Figure 12: 3D rendering of AM1(ASH) neurons.

Figure 13: 3D rendering of AM2(ADL) neurons.

Figure 14: 3D rendering of AM3(AWA) neurons.

Mentioned biomedical entities:

Genes: ADF, AM5, ASER, B, BAG, FLP, PHB, T12, URB, gcy-6, lsy-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, cell body, cilium, epidermal cell, neuron, secretory cell, sheath cell

Organisms: A. complexus, Acrobeles complexus, C. elegans, E. coli, H. contortus, Haemonchus contortus, Parastrongyloides trichosuri, swine

Tissues: anterior, cuticle, dorsal, dorsal nerve cord, excretory duct

Pathways: N/A

Query Result 5/11

Evolution of lateralized gustation in nematodes

Mackie, Marisa; Le, Vivian Vy; Carstensen, Heather R.; Kushnir, Nicole R.; Castro, Dylan L.; Dimov, Ivan M.; Quach, Kathleen T.; Cook, Steven J.; Hobert, Oliver; Chalasani, Sreekanth H.; Hong, Ray L.

bioRxiv

Year: 2025 PMID: 39282255 doi: 10.1101/2024.08.31.610597 Citations: 0 Score: 14.27

Keywords hits: ASE: 133, che-1: 80, lsy-6: 5

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in Pristionchus pacificus respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode Caenorhabditis elegans. By visualizing neuronal activity patterns, we show that contrary to previous expectations based on its genome sequence, the salt responses of P. pacificus are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Since NH4I sensation is affected by silencing of che-1 neurons but is unaffected in che-1 mutants, ASE function may be more greatly impacted by the silencing of ASE than by the loss of che-1. To assess whether Ppa ASE neurons show the same lateralized response to salt ions as Cel ASE neurons, we generated transgenic P. pacificus lines that express RCaMP in ASE neurons and assessed calcium responses to attractive salts. Specifically, although we observed weaker or comparable responses in AFD neurons when compared to either ASE neurons response towards 250 mM NH4 Cl, Na Cl and NH4I, the AFD responses were more robust than ASER toward 25 mM NH4 Cl, Na Cl, and NH4I. However, the responses to 250 mM Na Cl were not significantly reduced in the either the ASEL or ASER neuron in the gcy-22.3 mutant. We show that three neuron types each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. The worms were then incubated overnight at 37C with 500 l 4% -mercaptoethanol dissolved in 1% Triton-X100 in 0.1M Tris pH 7.4, and digested in 200 l of collagenase buffer with 200 units of collagenase type IV at 37C for 3.5 hours.

Figures:

Mentioned biomedical entities:

Genes: ADF, ASER, CHE-1, Cas9, Cel, HisCl1, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, fozi-1, gcy-5, lim-6, lin-4, lsy-6, rGC, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: neuron

Organisms: C. elegans, Pristionchus pacificus, human

Tissues: Sartorius

Pathways: N/A

Query Result 6/11

A Genome-Wide RNAi Screen for Factors Involved in Neuronal Specification in Caenorhabditis elegans, bScreen for Neuronal Fate Mutants

Poole, Richard J.; Bashllari, Enkelejda; Cochella, Luisa; Flowers, Eileen B.; Hobert, Oliver; Mango, Susan E.

PLoS Genetics

Year: 2011 PMID: 21698137 doi: 10.1371/journal.pgen.1002109 Citations: 0 Score: 13.58

Keywords hits: ASE: 353, che-1: 12, lsy-6: 35

Abstract:
One of the central goals of developmental neurobiology is to describe and understand the multi-tiered molecular events that control the progression of a fertilized egg to a terminally differentiated neuron. In the nematode Caenorhabditis elegans, the progression from egg to terminally differentiated neuron has been visually traced by lineage analysis. For example, the two gustatory neurons ASEL and ASER, a bilaterally symmetric neuron pair that is functionally lateralized, are generated from a fertilized egg through an invariant sequence of 11 cellular cleavages that occur stereotypically along specific cleavage planes. Molecular events that occur along this developmental pathway are only superficially understood. We take here an unbiased, genome-wide approach to identify genes that may act at any stage to ensure the correct differentiation of ASEL. Screening a genome-wide RNAi library that knocks-down 18,179 genes (94% of the genome), we identified 245 genes that affect the development of the ASEL neuron, such that the neuron is either not generated, its fate is converted to that of another cell, or cells from other lineage branches now adopt ASEL fate. We analyze in detail two factors that we identify from this screen: (1) the proneural gene hlh-14, which we find to be bilaterally expressed in the ASEL/R lineages despite their asymmetric lineage origins and which we find is required to generate neurons from several lineage branches including the ASE neurons, and (2) the COMPASS histone methyltransferase complex, which we find to be a critical embryonic inducer of ASEL/R asymmetry, acting upstream of the previously identified miRNA lsy-6. Our study represents the first comprehensive, genome-wide analysis of a single neuronal cell fate decision. The results of this analysis provide a starting point for future studies that will eventually lead to a more complete understanding of how individual neuronal cell types are generated from a single-cell embryo.

Summary:
In this paper we utilize the RNAi approach to screen, at a genome-wide level, for factors involved in all aspects of neuronal development. We are using the ASE gustatory neuron class, composed of two bilaterally symmetric neurons, ASEL and ASER, as a paradigm to understand how neuronal fate is specified. We find hlh-14 to be crucial for the specification of neurons that derive from several lineage branches, including the ASE neurons. Once a lineage is committed to only produce neurons, fates have to be differentially segregated to generate distinct types of neurons. How this bilateralization process is regulated at the levels of cell fate and cell morphogenesis is not known. The correct specification of the ASE neurons involves not only the imposition of ASE neuron identity that is shared by both the left and the right ASE neuron, but also involves the retention of the distinct lineage history of the two neurons. Since eri- lin-15b mutants generally seem less healthy than nre-1 lin-15b mutants, we chose the latter strain for our genome-wide screen. As the nre-1 lin-15b mutant background is relatively untested for its overall sensitivity to RNAi in general, we screened all RNAi clones on chromosome I not just for ASE neuronal phenotypes, but also for obviously visible phenotypes, and compared those to previously reported chromosome I screens of visible phenotypes, conducted either in a wildtype or an rrf-3 mutant background. More than 80% of genes in this category classify as genes involved in metabolism, cell/organelle structure and basic cellular processes like cell division. Interestingly, we were also able to identify from this tissue screen 21 genes that when knocked-down by RNAi led to a broad loss of neurons.

Figures:

Figure 1: The embryonic origins and specification of the ASE neurons.

Figure 2: Testing hypersensitive strains and validating the screen with visible phenotypes.

Figure 3: Overview of the screening strategy.

Figure 4: Examples of phenotypic classes uncovered by the screen.

Figure 5: Asymmetric expression and function ofhlh-14in the C-lineage.

Figure 6: The evolutionary conserved methyltransferase COMPASS complex is required for ASE asymmetric cell fate specification.

Figure 7: The cloning and expression oflsy-15/rbbp-5(ot86).

Mentioned biomedical entities:

Genes: ASER, Notch, lsy-6, mex-3

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, ectoderm, endoderm, mesoderm, neuron

Organisms: C. elegans

Tissues: nervous system, tissue

Pathways: N/A

Query Result 7/11

Single-cell transcriptional analysis of taste sensory neuron pair in Caenorhabditis elegans

Takayama, Jun; Faumont, Serge; Kunitomo, Hirofumi; Lockery, Shawn R.; Iino, Yuichi

Nucleic Acids Research

Year: 2010 PMID: 19875417 doi: 10.1093/nar/gkp868 Citations: 0 Score: 11.62

Keywords hits: ASE: 263, che-1: 17, lsy-6: 9

Abstract:

The nervous system is composed of a wide variety of neurons. A description of the transcriptional profiles of each neuron would yield enormous information about the molecular mechanisms that define morphological or functional characteristics. Here we show that RNA isolation from single neurons is feasible by using an optimized mRNA tagging method. This method extracts transcripts in the target cells by co-immunoprecipitation of the complexes of RNA and epitope-tagged poly(A) binding protein expressed specifically in the cells. With this method and genome-wide microarray, we compared the transcriptional profiles of two functionally different neurons in the main C. elegans gustatory neuron class ASE. Eight of the 13 known subtype-specific genes were successfully detected. Additionally, we identified nine novel genes including a receptor guanylyl cyclase, secreted proteins, a TRPC channel and uncharacterized genes conserved among nematodes, suggesting the two neurons are substantially different than previously thought. The expression of these novel genes was controlled by the previously known regulatory network for subtype differentiation. We also describe unique motif organization within individual gene groups classified by the expression patterns in ASE. Our study paves the way to the complete catalog of the expression profiles of individual C. elegans neurons.

Summary:

The primers used for the amplification of each gene were: lmn152: 5-CGTTCACCACCCACCAGAA-3, lmn132: 5-CAAGACGAGCTGATGGGTTATCT-3 for lmn-1 gcy551: 5-CCTACCAAGAGAAAAAGTTGAACTAAGAA-3, gcy531: 5-GGGCATGGATAGACCAACGA-3 for gcy-5 gcy651: 5-CACGTGGCGAAGTCATAATCA-3, gcy6-31: 5-GCTGACTCGTCCATTTTAAGCA-3 for gcy-6 gcy7. fw1: 5-TCTCCCAGACCCGATTTGG-3, gcy7. rv1: 5-CCGAAGGACGTTCGGTAAAAC-3 for gcy-7. Most of the genes expressed in ASEL and/or ASER have ASE motifs in their 5-upstream regions, which are specifically required for the expression in ASE neurons. We found that all of the novel ASER-biased genes had ASE motif in the promoter regions, whereas no ASE motif was found in the ASEL-biased ins-32 promoter. Therefore, we suggest that all of the eight novel ASER-biased genes could be directly regulated by CHE-1 binding to their ASE motifs whereas ins-32 expression is regulated indirectly. To determine whether the ASE motifs were enriched in the genes expressed in ASE neurons, we examined the frequency of ASE motifs per one kilobase promoter of each gene set categorized by the expression patterns in ASE. The frequency was higher in the genes expressed in either or both of ASE neurons than those not expressed in ASE neurons or those selected randomly from the microarray probe set, suggesting that the ASE motif is enriched in the genes expressed in ASE neurons.

Figures:

Figure 1: FLAG::PABP is expressed in single neurons of the transgenic animals.

Mentioned biomedical entities:

Genes: ASER, Bad, C14C6.3, CHE-1, L1, PABP, TRPC, cog-1, die-1, flp, fozi-1, gcy-22, gcy-5, gcy-6, ins-32, lim-6, lsy-6, nlp-5

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans

Tissues: C7, hypodermis, medial, nervous system

Pathways: N/A

Query Result 8/11

Plasticity in gustatory and nociceptive neurons controls decision making in C. elegans salt navigation

Dekkers, Martijn P. J.; Salfelder, Felix; Sanders, Tom; Umuerri, Oluwatoroti; Cohen, Netta; Jansen, Gert

Communications Biology

Year: 2021 PMID: 34504291 doi: 10.1038/s42003-021-02561-9 Citations: 0 Score: 9.92

Keywords hits: ASE: 290, che-1: 9, lsy-6: 2

Abstract:
A conventional understanding of perception assigns sensory organs the role of capturing the environment. Better sensors result in more accurate encoding of stimuli, allowing for cognitive processing downstream. Here we show that plasticity in sensory neurons mediates a behavioral switch in C. elegans between attraction to NaCl in nave animals and avoidance of NaCl in preconditioned animals, called gustatory plasticity. Ca2+ imaging in ASE and ASH NaCl sensing neurons reveals multiple cell-autonomous and distributed circuit adaptation mechanisms. A computational model quantitatively accounts for observed behaviors and reveals roles for sensory neurons in the control and modulation of motor behaviors, decision making and navigational strategy. Sensory adaptation dynamically alters the encoding of the environment. Rather than encoding the stimulus directly, therefore, we propose that these C. elegans sensors dynamically encode a context-dependent value of the stimulus. Our results demonstrate how adaptive sensory computation can directly control an animals behavioral state.

Summary:
This recovery of the response in ASEL is consistent with behavioral data, where attraction to Na Cl after pre-exposure is restored by a 5-min wash. We conclude that the ASEL neuron desensitizes with pre-exposure to Na Cl and recovers in the absence of Na Cl exposure and suggest that this sensory adaptation modulates the strength of attraction to Na Cl in behavioral assays. To test if the response of the ASH neurons is affected by pre-exposure, we first exposed animals to 100 mM Na Cl for 10 min and subsequently introduced a further 100 mM increase to 200 mM Na Cl. We also found that the ASER Ca2+ response likely involves a glutamate-mediated signal that advances the onset of the response, and a synaptic signal that abolishes a weak graded response in nave animals, neither of which are required for gustatory plasticity. Since ASE neurons are required for gustatory plasticity we tested whether they are required for sensitization of ASH by measuring ASH responses in che-1 mutants that lack functional ASE neurons. Thus, our computational model predicts that the robustness of the performance in the quadrant assay is not reduced by ASE sensitization. Next, we looked at the behavior of pre-exposed animals focusing on a 15 min 100 mM Na Cl pre-exposure, at which the avoidance behavior is the strongest.), confirming that in our model the separate roles of ASEL and ASER in motor control are direct consequences of the rise and decay times of their responses. Thus, our model results are consistent with results from ASEL and ASER ablated animals, as well as with observations that both steering and pirouettes contribute to the chemotaxis index in the quadrant assay. Accordingly, our model remains agnostic to these possibilities, as the encoding of food in the model is implicit. While multiple sensory neurons and contributions from the downstream circuitry most likely contribute to the rich behavioral responses in C. elegans, our simulations demonstrate the feasibility of a parsimonious model in which recruitment of ASH by an ASE derived signal underpins the switch between attraction and avoidance in gustatory plasticity. The bilaterally asymmetric kinetics of the ASE neurons, in which ASEL and ASER depolarize in response to concentration increases and decreases, respectively, suggest a potential to double the ASE sensory dynamic range, thus enhancing the resolution of Na Cl sensing in the animal.

Figures:

Figure 1: Ca2+responses of ASEL, ASER, and ASH neurons to brief NaCl exposure.

Figure 2: Prolonged exposure to NaCl sensitizes ASER.

Figure 3: Pre-exposure to NaCl desensitizes ASEL.

Figure 4: Prolonged exposure to NaCl sensitizes ASH.

Figure 5: ASER sensitization is affected byunc-13andeat-4mutations.

Figure 6: ASH sensitization requires ASE, glutamate, neuropeptides, dopamine, and serotonin.

Figure 7: Computational model reproduces sensory responses to NaCl in virtual quadrant assay.

Figure 8: Stochastic recruitment of ASH drives gustatory plasticity in our computational model.

Figure 9: Sensory adaptation controls exploration in a virtual spot assay.

Figure 10: Schematic model of the NaCl navigation circuit.

Mentioned biomedical entities:

Genes: ADF, ASER, CFP, lsy-6, sra-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, interneuron, neuron

Organisms: C. elegans

Tissues: capillary, nervous system

Pathways: N/A

Query Result 9/11

microRNA function in left-right neuronal asymmetry: perspectives from C. elegans

Alqadah, Amel; Hsieh, Yi-Wen; Chuang, Chiou-Fen

Frontiers in Cellular Neuroscience

Year: 2013 PMID: 24065887 doi: 10.3389/fncel.2013.00158 Citations: 0 Score: 8.43

Keywords hits: ASE: 62, che-1: 3, lsy-6: 20

Abstract:
Leftright asymmetry in anatomical structures and functions of the nervous system is present throughout the animal kingdom. For example, language centers are localized in the left side of the human brain, while spatial recognition functions are found in the right hemisphere in the majority of the population. Disruption of asymmetry in the nervous system is correlated with neurological disorders. Although anatomical and functional asymmetries are observed in mammalian nervous systems, it has been a challenge to identify the molecular basis of these asymmetries. C. elegans has emerged as a prime model organism to investigate molecular asymmetries in the nervous system, as it has been shown to display functional asymmetries clearly correlated to asymmetric distribution and regulation of biologically relevant molecules. Small non-coding RNAs have been recently implicated in various aspects of neural development. Here, we review cases in which microRNAs are crucial for establishing leftright asymmetries in the C. elegans nervous system. These studies may provide insight into how molecular and functional asymmetries are established in the human brain.

Summary:
The hypothesis that differential expression of these genes between left and right sides of the cortex may be regulated by miRNAs is plausible and worth further investigation. Although there are limited reports of the involvement of miRNAs in the development of neuronal asymmetry in vertebrates, C. elegans has proved to be a powerful model organism to study lateralization of the nervous system due to its genetic amenability, as well as evidence of functional asymmetry having clear molecular correlates. Both involve the specification of two types of chemosensory neurons: the pair of amphid neurons, single cilliated endings taste neurons, in which lsy-6 and mir-273 miRNAs are involved, and the pair of amphid wing C olfactory neurons, where mir-71 is crucial for establishment of asymmetry. Like other animals, the C. elegans nervous system appears generally symmetric. However, the pair of taste neurons, called ASE left and ASE right displays molecular and functional asymmetries. In the ASEL neuron, lsy-6 directly represses an ASER promoting transcription factor COG-1, through physical binding of complementary bases in the cog-1 3UTR. However, AWC left and AWC right neurons express different odorant receptors and sense different odors.

Figures:
Figure 1: Priming and boosting model of ASE asymmetry.

Figure 2: Model of mir-71 function in AWC asymmetry.

Mentioned biomedical entities:
Genes: ASER, COG-1, DIE-1, LIN-14, LMO4, Myt1, Notch, TRIM32, die-1, lin-4, lsy-6, mir-273, mir-71, nsy-4, str-2, xrn-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron, progenitor cell

Organisms: C. elegans, humans, mice, zebrafish

Tissues: epithalamus, nervous system

Pathways: N/A

Query Result 10/11

Cell-type specific sequencing of microRNAs from complex animal tissues

Alberti, Chiara; Manzenreither, Raphael A.; Sowemimo, Ivica; Burkard, Thomas R.; Wang, Jingkui; Mahofsky, Katharina; Ameres, Stefan L.; Cochella, Luisa

Nature methods

Year: 2018 PMID: 29481550 doi: 10.1038/nmeth.4610 Citations: 0 Score: 6.28

Keywords hits: ASE: 57, che-1: 2, lsy-6: 5

Abstract:

MicroRNAs (miRNAs) play an essential role in the post-transcriptional regulation of animal development and physiology. However, in vivo studies aimed at linking miRNA-function to the biology of distinct cell types within complex tissues remain challenging, partly because in vivo miRNA-profiling methods lack cellular resolution. We report microRNome by methylation-dependent sequencing (mime-seq), an in vivo enzymatic small RNA-tagging approach that enables high-throughput sequencing of tissue- and cell-type-specific miRNAs in animals. The method combines cell-type-specific 3-terminal 2-O-methylation of animal miRNAs by a genetically encoded, plant-specific methyltransferase (HEN1), with chemoselective small RNA cloning and high-throughput sequencing. We show that mime-seq uncovers the miRNomes of specific cells within C. elegans and Drosophila at unprecedented specificity and sensitivity, enabling miRNA profiling with single-cell resolution in whole animals. Mime-seq overcomes current challenges in cell-type-specific small RNA profiling and provides novel entry points for understanding the function of miRNAs in spatially restricted physiological settings.

Summary:

Moreover, worms and flies expressing ubiquitous Ath-HEN1 were viable and fertile and showed no signs of altered miRNA function. To assess the sensitivity and specificity of methylation-dependent small RNA cloning and sequencing, we diluted total RNA from Drosophila S2-hen1KO cells stably expressing Ath-HEN1 with total RNA from mouse embryonic stem cells, in ratios spanning five orders of magnitude. This lower overall recovery was expected, as miR strands are typically low in abundance because of their rapid decay upon loading of the partner miR strand onto Argonaute,. Our results suggest that directing Ath-HEN1 expression to specific cells should enable recovery of the most relevant miRNAs expressed in a single cell out of whole C. elegans worms at a TPR of 0.88, and even in single neurons within fly brains at a TPR of 0.80, in both cases with very low FPRs. To test if Ath-HEN1-directed methylation followed by oxidation-resistant cloning preserves the relative quantity of miRNAs, we compared the abundance of untreated Drosophila S2 miRNAs to that of the Drosophila miRNAs that are significantly enriched after oxidation, throughout the S2-Ath-HEN1/mESC dilution series. From a total of 40 reporters, only four miRNAs showed neuronal expression but were not significantly enriched by mime-seq. In direct comparison with neuronal and intestinal Argonaute immunoprecipitations in C. elegans,, mime-seq identified 3- and 2.6-fold more miRNAs respectively, many of which we validated, including a miRNA expressed in a single neuron out of the whole animal. Comparison of the miRNomes obtained by mime-seq with transcriptional reporters for several C. elegans miRNAs showed a high correspondence, both for enriched and depleted miRNAs. Note, that one expressed miRNA aligned to both the mouse and the Drosophila genome but was assigned to the Drosophila sample because miR-184 is only spuriously expressed in mESC. The Nextflow pipeline developed for this analysis, as well as the R script used for further analysis can be found at: https://gitlab. com/tburk/smallRNA-meth To test the enrichment of Drosophila miRNAs in the titration experiment in, we considered Drosophila and mouse miR strands with an average cpm 10 over the three replicates for untreated samples with 1:1 dilutions and also their associated miR strands.

Figures:

Figure 1: The methyltransferase HEN1 fromArabidopsis thalianamethylates animal miRNAsin vitroandin vivo.

Figure 2: Periodate-mediated oxidation enables selective cloning of animal miRNAs upon restrictive Ath-HEN1-directed methylation.

Figure 3: Mime-seq reproducibly recovers neuronal miRNAs inC. elegans.

Figure 4: Mime-seq unveils the miRNomes of diverse tissues inC. elegans.

Figure 5: Mime-seq has the sensitivity to retrieve miRNAs from 1/300 cells.

Mentioned biomedical entities:

Genes: Actin, Actin5C, Argonaute, Cas9, Gal4, HEN1, HENN-1, Hen1, RDE-1, Tubulin, Unc, attB, lsy-6, miR-252, miR-277, rab-3, rgef-1, unc-31

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, Mus musculus, mouse

Tissues: cuticle, seed, tissue, tube

Pathways: N/A

Query Result 11/11

MicroRNAs: Not Fine-Tuners but Key Regulators of Neuronal Development and Function

Davis, Gregory M.; Haas, Matilda A.; Pocock, Roger

Frontiers in Neurology

Year: 2015 PMID: 26635721 doi: 10.3389/fneur.2015.00245 Citations: 0 Score: 5.57

Keywords hits: ASE: 63, che-1: 1, lsy-6: 7

Abstract:

MicroRNAs (miRNAs) are a class of short non-coding RNAs that operate as prominent post-transcriptional regulators of eukaryotic gene expression. miRNAs are abundantly expressed in the brain of most animals and exert diverse roles. The anatomical and functional complexity of the brain requires the precise coordination of multilayered gene regulatory networks. The flexibility, speed, and reversibility of miRNA function provide precise temporal and spatial gene regulatory capabilities that are crucial for the correct functioning of the brain. Studies have shown that the underlying molecular mechanisms controlled by miRNAs in the nervous systems of invertebrate and vertebrate models are remarkably conserved in humans. We endeavor to provide insight into the roles of miRNAs in the nervous systems of these model organisms and discuss how such information may be used to inform regarding diseases of the human brain.

Summary:

Prior to the discovery of the miRNA pathway, the lin-14 gene in Caenorhabditis elegans was shown to be regulated by a 22-nucleotide partially complementary strand of RNA called lin-4. Mir-1 also targets synaptic proteins neuroligin and neurexin, which in humans are two synaptic proteins that have been linked with defects in synaptic function associated with autism spectrum disorders. For example, control of neuronal progenitor proliferation is fine-tuned by the highly conserved miRNA, miR-124, which has been shown in various organisms to regulate neuronal stem cells. Deletion of Dicer leads to neuron migration defects and impaired cellular differentiation, as well as cell fate changes and cortical lamination defects. Conditional deletion of Dicer from the embryonic day 8 telencephalon causes a loss of radial glial progenitor markers, including nestin, Sox9, and ErbB2, which then results in an increase in basal progenitors and postmitotic neurons. Defects associated not only with the expression of key miRNAs, but also at a genetic level have been implicated in schizophrenia.

Figures:

Figure 1: The miRNA pathway .

Figure 2: Roles of miRNAs in different stages of neuronal development .

Mentioned biomedical entities:

Genes: AGO, ASCL1, ASER, Akt, Argonaute, CHE-1, COG-1, CamkII, DAF-16, Dgcr8, Dicer, Drosha, EGFR, Emx1, ErbB2, FMR1, FOXO, Fgf, Fgf8, Fmr1, Foxg1, Foxp2, Gsh2, Hes1, LIN-14, LIN-41, Let-7, Lmx1b, MAP2, MECP2, MEF-2, MLN51, MYT1L, MeCP2, NEUROD2, NRXN1, Notch, Nrg, Pasha, Phf6, RNA, RhoG, SHANK3, SOX-2, SQV-5, Sox9, Syt1, TLX, TRIM, TRIM32, UNC-40, Wnt, XPO5, ana, chinmo, dicer, fgfr1, her5, let-7, lin-14, lin-28, lin-4, lin-41, lsy-6, miR-124, miR-125, miR-125b, miR-137, miR-228, miR-289, miR-7, miR-9, miR-92b, miR-9a, mir-71, mir-79, pitx3, zic5

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: axon, cell, glial cell, neural progenitor cell, neural stem cell, neuron, stem cell

Organisms: C. elegans, human, humans, mice, mouse, zebrafish

Tissues: epidermis, nervous system, neural crest, neural tube, neuromuscular junction, tissue

Pathways: N/A